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CMSC 207 CRN 32950

CMSC 207: Chapter 9 Homework

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1. Prove that for all integers n, k, and r for $r \leq k \leq n$ that:

$$\binom{n}{k} \cdot \binom{k}{r} = \binom{n}{r} \cdot \binom{n-r}{k-r}.$$

2. The binomial theorem states that for any numbers a and b:

$$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k} b^k \text{ for any integer } n \geq 0.$$

Use this theorem to show that for any integer $n \geq 0$:

$$\sum_{k=0}^{n} (-1)^k \binom{n}{k} 3^{n-k} 2^k = 1.$$